GEOAI • ENVIRONMENTAL INTELLIGENCE

GeoPilot AI

An AI-assisted outdoor activity planning prototype using weather forecasts and air-quality indicators.

WHAT IT DOES
Evaluates hourly conditions and explains environmental risks before an outdoor activity.

HOW IT WORKS
Location lookup → hourly weather and air quality → heuristic ranking → AI-assisted explanation.

DESIGN PRINCIPLE
Recommendations communicate risk clearly; they do not promise that conditions are safe.

Prototype status: core API and interface implemented; external connectivity and public deployment still require validation.

01 / THE PROBLEM

Outdoor decisions need more than a weather icon

People planning a run, walk or other outdoor activity may overlook pollution exposure or changing hourly conditions. A single temperature or rain forecast cannot represent the full context.

WEATHER Temperature, precipitation probability, rainfall and wind can vary by hour.
AIR QUALITY AQI and PM2.5 indicators can reveal an environmental concern even when the sky is clear.
PERSONAL CONTEXT Activity type and user-stated constraints influence how the prototype ranks conditions.

Goal: turn environmental data into an understandable activity-planning aid, with explicit warnings when conditions are unfavorable.

02 / SYSTEM DESIGN

A lightweight, modular pipeline

1. LOCATION Geocoding API resolves the requested place to coordinates.	**2. ENVIRONMENT** Hourly forecast and air-quality APIs provide weather, AQI and PM2.5 inputs.	**3. RANKING** A prototype heuristic scores available hours and records risk factors.	**4. EXPLANATION** Local Qwen2.5-1.5B can generate an explanation; a fallback is available.

CURRENT STACK

Python 3.12	FastAPI	Streamlit	Ollama + Qwen2.5-1.5B
Open-Meteo Geocoding	Open-Meteo Forecast	Open-Meteo Air Quality	HTTP API integration

The prototype exposes data sources and limitations rather than presenting heuristic scores as validated safety guidance.

Risk-aware recommendations, not blind optimism

In a prior successful API test, the system evaluated a Bhubaneswar jogging request and identified elevated forecast AQI and PM2.5 as reasons for strong caution.

160	73.8 µg/m³	3 / 100
Forecast US AQI	Forecast PM2.5	Prototype score

Advisory returned: strong caution; the prototype advised against outdoor exertion for that forecast window. This was a test result, not a claim about current conditions.

IMPORTANT LIMITATIONS

- Scores are heuristic and not clinically or operationally validated.
- External Open-Meteo access failed during a later test due to DNS resolution, so live API reliability needs improvement.
- The local API is not yet public; it cannot be evaluated remotely while running only on localhost.

From prototype to geospatial decision support

NOW
Core API, hourly forecast evaluation, pollution warnings, local model integration and Streamlit UI.

NEXT
Deploy a public API; add robust network error handling; publish repository and model documentation; test request examples.

FUTURE
OpenStreetMap nearby features, road-network context, green cover, water bodies and satellite-derived layers—after implementation and validation.

DATA SOURCES CURRENTLY USED

Open-Meteo Geocoding API • Open-Meteo Forecast API • Open-Meteo Air Quality API (when available).

Open-weight language model: Qwen2.5-1.5B via Ollama. The final submission still needs an accessible public model card/link.

Submission status: presentation draft. Public API deployment, public repository and final link verification remain required.